This overview details how data quality evolved from the initial raw exports to the final cleaned datasets. Detailing the path from raw data to cleaning decisions and final outcomes, this approach ensures corrections were made only when fully supported by the source data, while unresolvable anomalies were safely preserved, nulled, or flagged for review.

1.1 Key Quality Improvements

Key data quality improvements performed during the data cleaning:

- Standardization of text and language data: Across tables (Managers, Courses, Cohorts), mixed languages were resolved, and employee and teacher names were standardized and transliterated into English.
- Handling of duplicates: Duplicate rows were identified and removed in key tables, including Leads where duplicate LeadIDs were resolved, MarketingSpend where 79 total duplicates were removed, Payments where 74 total duplicates were removed, and StudentActivity where 567 total duplicates were removed.
- Missing values and imputation: In the MarketingSpend table, missing campaign values were filled with 'not set' and missing spend values were filled with 0. In the Payments table, payment dates for refunded rows were filled and all number fields were mapped to correct number types using the clean_money function.
- Validation of logical and numerical anomalies: In the StudentActivity table, invalid numeric values and anomalies were processed, including negative activity values, quiz scores greater than 100 (627 rows), cases where homework accepted exceeded homework submitted (642 rows), and future activity dates (177 rows). Invalid numeric metrics were converted to null.
- Financial and status flagging: In the Payments, Enrollments, and StudentActivity tables, validation fields and flags were added to correctly track statuses such as cancelled and certified enrollments, and refund details.

1.2 The following data characteristics and records were intentionally preserved or specifically flagged during data processing to maintain analytical integrity and business logic:

- Legitimate Null Termination Dates: In the Managers dataset, null values in the termination date column for currently active employees were intentionally preserved rather than arbitrarily filled.
- Full Retention of Course Records: All 17 records in the Courses dataset were fully retained with zero deletions, ensuring complete baseline coverage.
- Enrollment Status Flagging: No rows were removed from the Enrollments dataset; instead, cancelled and certified enrollments were explicitly categorized and flagged using dedicated validation fields (is cancelled and is_certified).
- Exchange Rate Validation and Retention: All 3,644 rows in the Exchange Rates dataset were retained, with positive rate values and the baseline UAH rate of 1 fully validated and preserved.
- Full Cohort Records Retention: All 242 records in the Cohorts dataset were retained without dropping any entries, keeping the structural timeline intact.

1.3 Final Data Quality Status

The final data quality status of all datasets after cleaning, validation, and preparation:

- Managers: All 8 raw rows were retained with no duplicates, unique ManagerID, standardized English names, and legitimate null termination dates preserved.

- Courses: All 17 raw rows were retained with unique CourseID, validated rules/currencies, and transliterated English teacher names.
- Leads: Cleaned down from 51,779 raw rows to 51,021 unique LeadID values, resolving duplicate IDs and standardizing campaign values.
- Marketing Spend: Cleaned down from 13,968 raw rows to 13,889 rows after removing total duplicates, recovering spend values, filling missing campaign values with 'not set', and filling missing spend with 0.
- Payments: Cleaned down from 17,174 raw rows to 17,100 rows after removing duplicates, filling payment/refund dates, mapping number fields via clean_money, and adding validation fields.
- Enrollments: All 12,161 raw rows were retained with unique StudentID, zero duplicates, and dedicated flags for cancelled and certified statuses.
- Student Activity: Cleaned down from 111,641 raw rows to 111,074 rows after removing duplicates, converting invalid numeric values and anomalies to null, and adding validation fields.
- Exchange Rates: All 3,644 rows were retained with no missing values, positive rates, and a validated UAH rate of 1.
- Cohorts: All 242 records were retained with unique CohortID, correct data types, and standardized English teacher names.

• Dataset	Raw Rows	Cleaned Rows	Main Initial Issues	Final State
Managers	8	8	Five null TerminationDate values; mixed data	All 8 records

• Dataset	Raw Rows	Cleaned Rows	Main Initial Issues	Final State
			types; mixed/malformed date formats; Mixed languages in column ManagerName	remained (no duplicates); unique ManagerID; legitimate termination-date nulls preserved; All columns mapped to correct data types; Standardized and transliterated ManagerNames into a unified script in English
Courses	17	17	No material data-quality issue identified; no missing values; no duplicates; mixed data types;mixed/malformed date formats; Mixed languages in column TeacherName	All 17 course records remained; unique CourseID; dates, numeric rules and currency validated; All columns mapped to correct data types; Standardized and transliterated TeacherNames into a unified script in English
Leads	51,779	51,021	Duplicate LeadID records, missing values in columns **FirstContactAt, ConvertedAt, FullName, Region, City, PreferredLanguage,LeadScore, Campaign, Content, Term, ReferrerDomain;** mixed data types;mixed/malformed date formats; inconsistent (casing/invalid/blank/alias) contact(email,phone)/geography(city,region,country/source/language fields, mixed dates and	51,021 unique LeadID values; issues with duplicate IDs were resolved; cleaned and validation fields added;

• Dataset	Raw Rows	Cleaned Rows	Main Initial Issues	Final State
			invalid expected prices; Mixed languages in column FullName, ManagerNameRaw, email;	
MarketingSpend	13,968	13,889	79 total duplicates, missing Spend, campaign variants, spend/base-currency mismatches and click/impression anomalies;	duplicates dropped; campaign values standardized; 79 missing spend values recovered; missed values in Campaign is filled with 'not set', missed values in Spend column is filled with 0
Payments	17,174	17,100	74 total duplicates, missing values in columns **PaymentDate, RefundDate** , **RefundReason, InvoiceNumber;** mixed data types;mixed/malformed date formats; wrong codes in column currency, 58 rows with ['GrossAmount'] < ['RefundAmount']; 112 rows ['InstallmentNumber'] > ['InstallmentCount'];	duplicates removed by column [Payment*ID]*; flagged by column "is_refunded"; PaymentDates are filled with RefundDate;all number fields are mapped to correct number types using function clean_money; cleaned and validation fields added;
Enrollments	12,161	12,161	missing values in columns ActualCompletionDate, CancellationDate, CancellationReason and CertificateIssuedDate; mixed data types;mixed/malformed date formats;	No rows removed; unique EnrollmentID; no duplicates;

• Dataset	Raw Rows	Cleaned Rows	Main Initial Issues	Final State
				cancelled and certified enrollments are flagged ‘is_cancelled’ and ‘is_certified’
StudentActivity	111,641	111,074	567 total duplicates, negative activity values, invalid quiz scores, homework/date anomalies and weekly gaps; mixed data types;mixed/malformed date formats; 642 rows with ['HomeworkAccepted'] >['HomeworkSubmitted']; 627 rows with ['AverageQuizScore'] > 100; 125 rows with ['LessonsCompleted'] >['LessonsViewed']; 110 rows with Logins < 0; 107 rows with ['VideoMinutesWatched'] < 0; 177 rows with ['LastActivityAt'] > today	duplicates removed; invalid numeric values converted to null; cleaned and validation fields added;
ExchangeRates	3,644	3,644	No missing values;	All rows retained; positive rates and UAH rate of 1 validated;
Cohorts	242	242	No missing values; mixed data types;mixed/malformed date formats; Mixed languages in column TeacherName	All 242 records retained (no duplicates); unique CohortID; All columns mapped to correct data types; course, Standardized and transliterated TeacherNames into a unified script in English